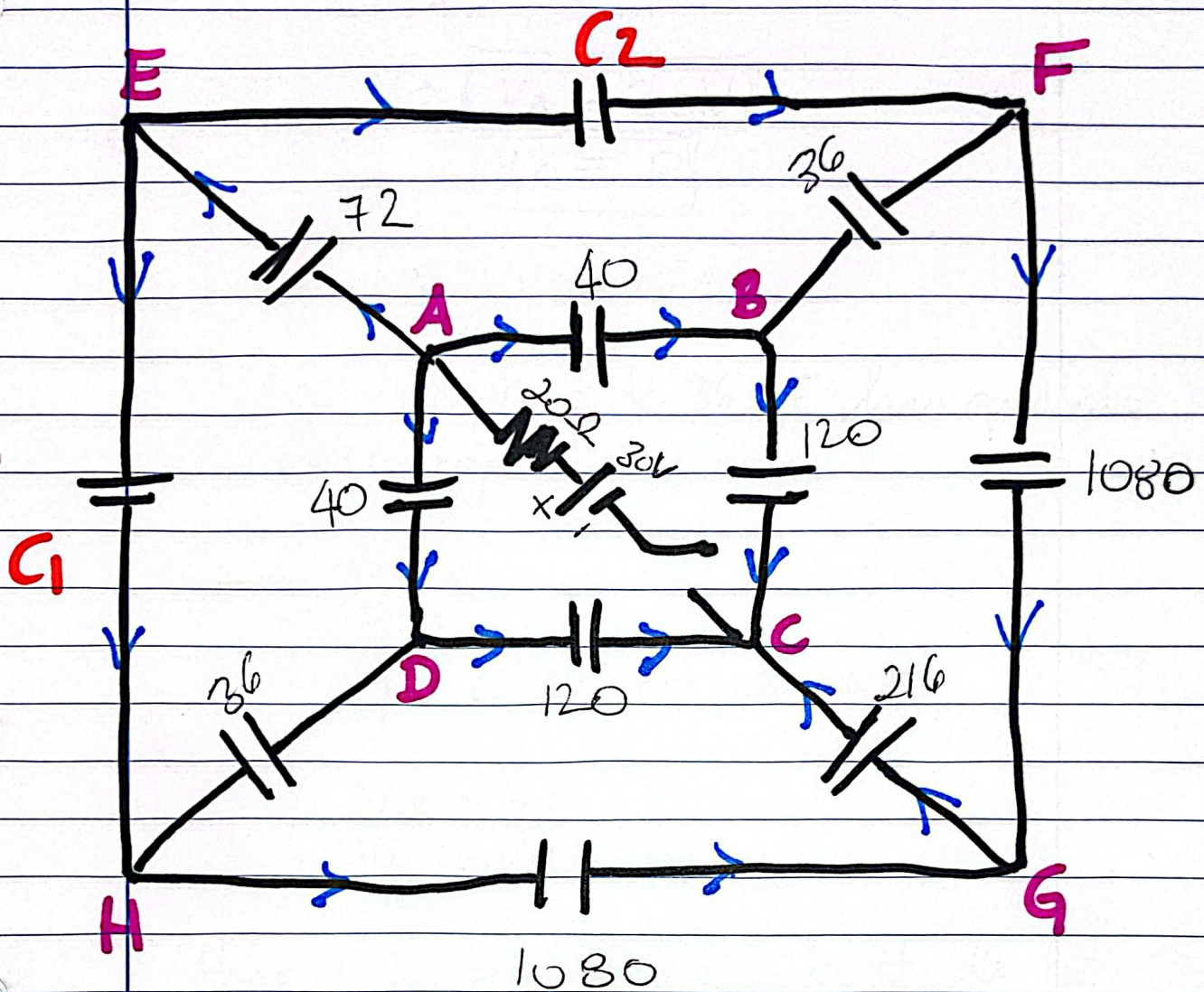


Problema #1. Capacitores



cen μF .

a)

$$\Delta V_{AC} = 30V.$$

$$\hookrightarrow \begin{cases} V_A = 30V \\ V_C = 0V \end{cases}$$

Luego, el problema dice:

"los capacitores de $36 \mu F$ han quedado descargados".

$$\rightarrow q = 0$$

$$q = C \Delta V. \rightarrow \Delta V = 0$$

$$\begin{cases} V_B = V_F \\ V_D = V_H \end{cases}$$

Ahora, la ley de Kirchhoff de las corrientes

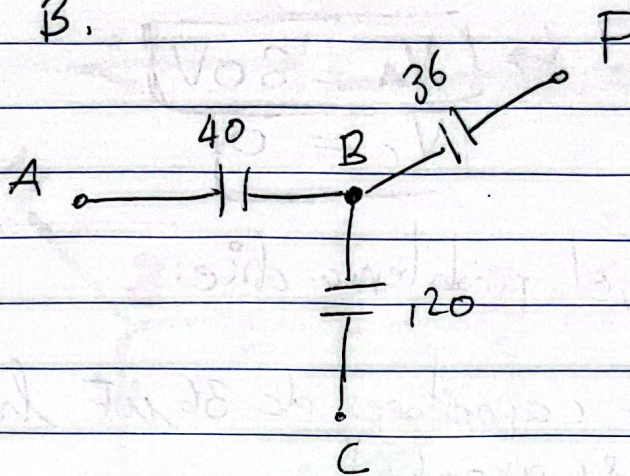
$$\sum I = 0, \text{ para los nodos}$$

Cuando ya los ~~cargados~~ capacitores están cargados, ya no hay corriente pero sirve con capacitores.

$$\sum C \Delta V = 0$$

Enfoces, des puds de un tiempo:

• Nodo B.



Kirchhoff:

$$40(V_B - 30) + 36(V_B - V_F) + 120(V_B - V_C) = 0$$

$$\Leftrightarrow 40(V_B - 30) + 120V_B = 0$$

$$\boxed{V_B = 7,5V}, \quad \boxed{V_F = 7,5V}$$

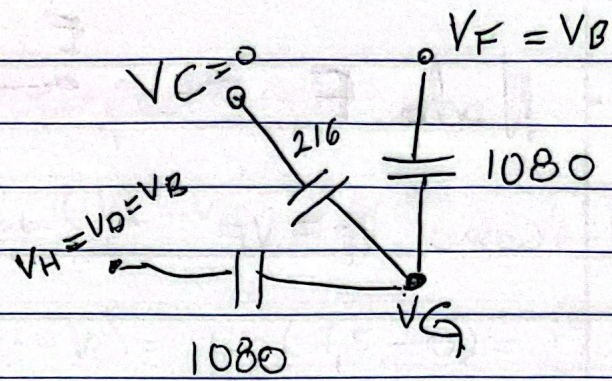
• Nodo D.

Como el prob. es simétrico con respecto al eje E-G.

$$V_D = V_B$$

$$\boxed{V_D = 7,5V}, \quad \boxed{V_H = 7,5V}$$

• Nodo G

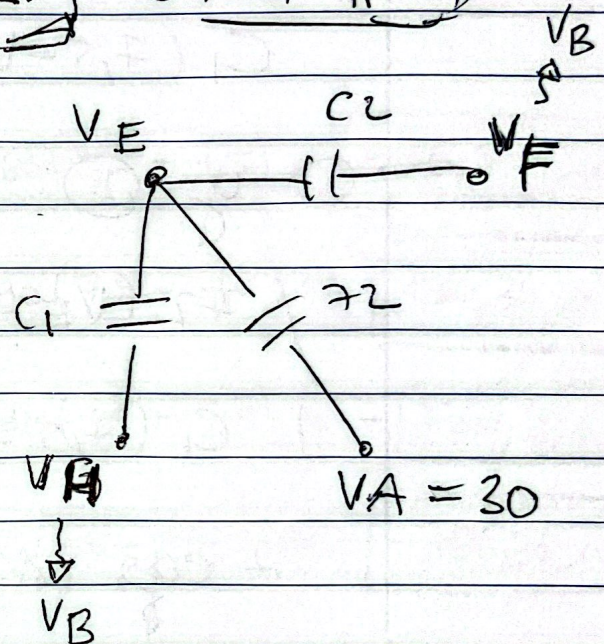


$$\sum C(\Delta V) = 0.$$

$$1080(V_G - V_B) + 216(V_G) + 1080(V_G - V_B) = 0$$

$$\rightarrow \boxed{V_G = 6,82V} \quad \text{ó} \quad \boxed{V_G = \frac{10}{11}V_B}$$

• Nodo E



$$\sum C(\Delta V) = 0$$

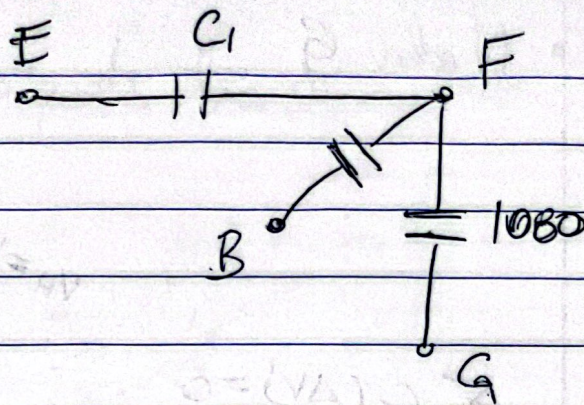
$$72(V_E - 30) + C_1(V_E - V_B) + C_2(V_E - V_B) = 0$$

por simetría, $\boxed{C_1 = C_2}$

$$72(V_E - 30) + 2C_1(V_E - V_B) = 0$$

(1)

Nodo F



Como $V_B = V_F$

$$C_1 (V_B - V_E) + 1080 (V_B - V_G) = 0$$

$$\Rightarrow C_1 (7,5 - V_E) + 1080 \left(7,5 - \frac{10}{11} 17,5 \right) = 0$$

$$C_1 (7,5 - V_E) + \frac{8100}{11} = 0 \quad (2)$$

① y ②:

$$\begin{cases} 72 (V_E - 30) + 2C_1 (V_E - 7,5) = 0 \\ C_1 (7,5 - V_E) + 8100/11 = 0 \end{cases}$$

2 eqs con 2 incógnitas

$$\boxed{V_E \approx 9,55 \text{ V}} = \boxed{\frac{105}{11} \text{ V}}$$

$$\boxed{C_1 = C_2 = 360 \mu\text{F}}$$

b) Carga en capacitores

$$Q_{AB}: C_{ab}(V_A - V_B) = 40(30 - 7,5) = \boxed{900 \mu C}$$

$$Q_{BC}: C_{bc}(V_B - V_C) = 120(7,5 - 0) = \boxed{900 \mu C}$$

$$Q_{CD}: C_{cd}(V_C - V_D) = 120(0 - 7,5) = \boxed{-900 \mu C}$$

$$Q_{DA}: C_{da}(V_D - V_A) = 40(7,5 - 30) = \boxed{-900 \mu C}$$

$$Q_{EF}: C_{ef}(V_E - V_F) = 360\left(\frac{105}{11} - 7,5\right) \approx \boxed{736,36 \mu C}$$

$$Q_{FG}: C_{fg}(V_F - V_G) = 1080\left(7,5 - \frac{75}{11}\right) \approx \boxed{736,36 \mu C}$$

$$Q_{GH}: C_{gh}(V_G - V_H) = 1080\left(\frac{75}{11} - 7,5\right) \approx \boxed{-736,36 \mu C}$$

$$Q_{HE}: C_{he}(V_H - V_E) \approx 360\left(7,5 - \frac{105}{11}\right) \approx \boxed{-736,36 \mu C}$$

$$Q_{AE} = C_{ae}(V_A - V_E) = 72\left(30 - \frac{105}{11}\right) \approx \boxed{1472,73 \mu C}$$

$$Q_{BF} = C_{bf}(V_B - V_F) = 36(7,5 - 7,5) = 0$$

$$Q_{CG} = C_{cg}(V_C - V_G) = 216\left(0 - \frac{75}{11}\right) \approx \boxed{-1472,73 \mu C}$$

$$Q_{DH} = C_{dh}(V_D - V_H) = 36(7,5 - 7,5) = 0$$

c) Energía total en los capacitores

$$E_T = \frac{1}{2} \sum C (\Delta V)^2$$

$$\left(\frac{1}{2}\right)(40)(30 - 7,5)^2$$

$$+ \left(\frac{1}{2}\right)(120)(7,5 - 0)^2$$

$$+ \left(\frac{1}{2}\right)(120)(0 - 7,5)^2$$

$$+ \left(\frac{1}{2}\right)(40)(7,5 - 30)^2$$

$$+ \frac{1}{2}(360)\left(\frac{105}{11} - 7,5\right)^2$$

$$+ \left(\frac{1}{2}\right)(1080)\left(7,5 - \frac{75}{11}\right)^2$$

$$+ \left(\frac{1}{2}\right)(1080)\left(\frac{75}{11} - 7,5\right)^2$$

$$+ \left(\frac{1}{2}\right)(360)\left(7,5 - \frac{105}{11}\right)^2$$

$$+ \left(\frac{1}{2}\right)(72)\left(30 - \frac{105}{11}\right)^2$$

$$+ \left(\frac{1}{2}\right)(216)\left(0 - \frac{75}{11}\right)^2$$

$$* = 49\,090 \text{ } \mu\text{J}$$

d) Energía entregada por la batería:

$$W_{\text{Batería}} = \int_0^{\infty} V_{\text{bat}} i(t) dt$$

$$= V_{\text{bat}} \int_0^{\infty} i(t) dt$$

$$= V_{\text{bat}} \cdot Q$$

donde Q es la que sale del nodo A.

$$Q = Q_{AE} + Q_{AD} + Q_{AB}$$

De la parte b),

$$Q = 1472,73 + 900 + 900 = 3272,73 \mu\text{C}$$

Entonces

$$W_b = V_b Q_T \\ = (30\text{V}) (3272,73) = \boxed{98,18 \text{ mJ}}$$